Reproducible Performance Measurement and Analysis for High-Performance Computing Applications

HPDC Tutorial

July 20, 2025

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Tutorial Presenters

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LLNL

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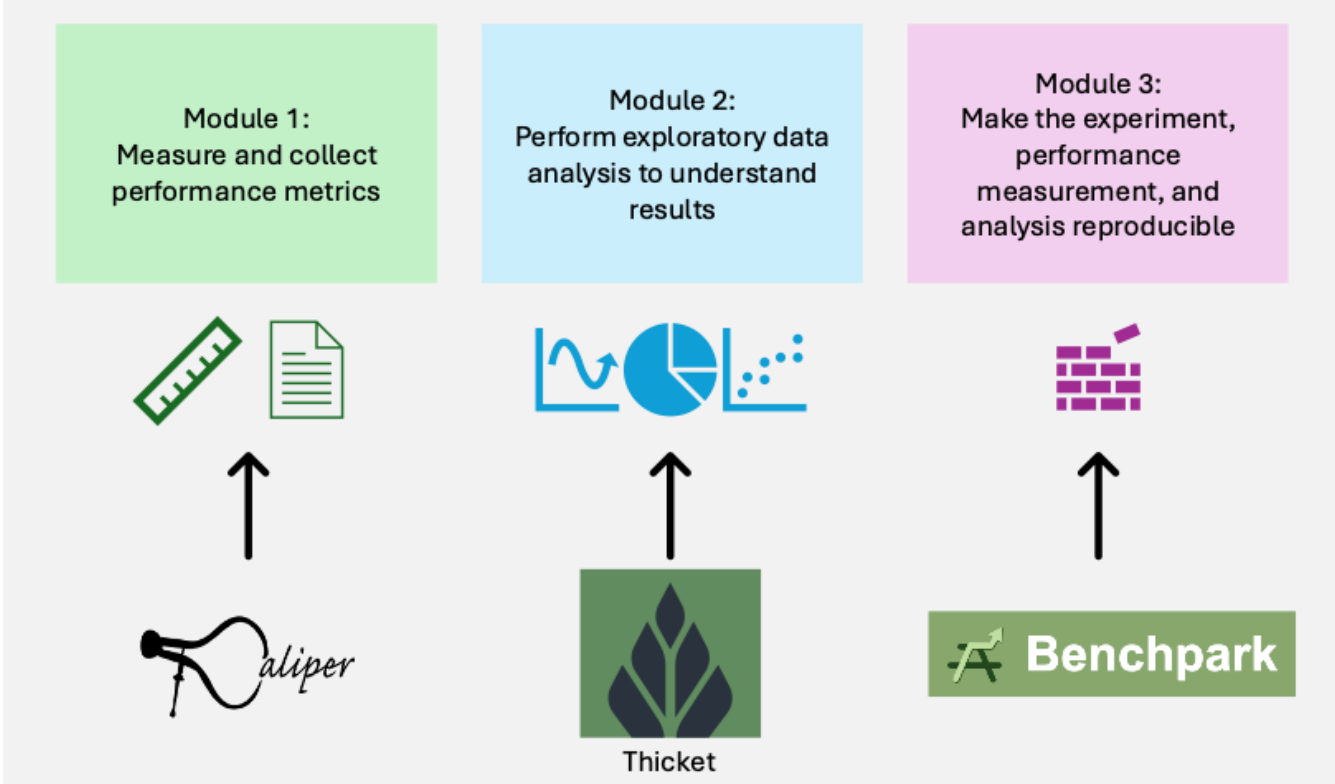
UTK

Overall Tutorial Agenda (approximate)

Welcome and Introduction	15 min
Module 1: Performance measurement with Caliper	45 min
Module 2: Performance analysis with Thicket	30 min
Break	30 min
Module 2 (cont'd): Performance analysis with Thicket	20 min
Module 3: Reproducible experiments with Benchpark	60 min
Q&A and Wrapup	10 min



*Hands on component to accompany each module

Overall Tutorial Materials

Find these slides and associated scripts here:

- Caliper: https://software.llnl.gov/Caliper/ under “Materials”
- Thicket: https://thicket.readthedocs.io/en/latest/tutorial_materials.html
- Benchpark: https://software.llnl.gov/benchpark/example-workflow.html

We also have a channel on Spack slack.

You can join here:

slack.spack.io

Join the **#benchpark-support** channel!

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Tutorial Instances: http://bit.ly/4kGQDlc

- We have an AWS instance for the hands-on component of this tutorial
- The instance provides:
  - Pre-installed Thicket, Caliper, and Benchpark and required dependencies
  - Caliper source code demos
  - Thicket Jupyter notebooks and datasets for performance analysis

When logging in to the instance:

- Please use a unique username to avoid resource allocation conflicts
  - First initial followed by last name (e.g., jdoe)
- PW: hpctutorial25

Performance Measurement with Caliper

David Boehme

Performance Analysis with Thicket I

Stephanie Brink

30 Min Break

Performance Analysis with Thicket II

Stephanie Brink

Reproducible Experiments with Benchpark

Stephanie Brink

We benchmark HPC systems for a variety of reasons

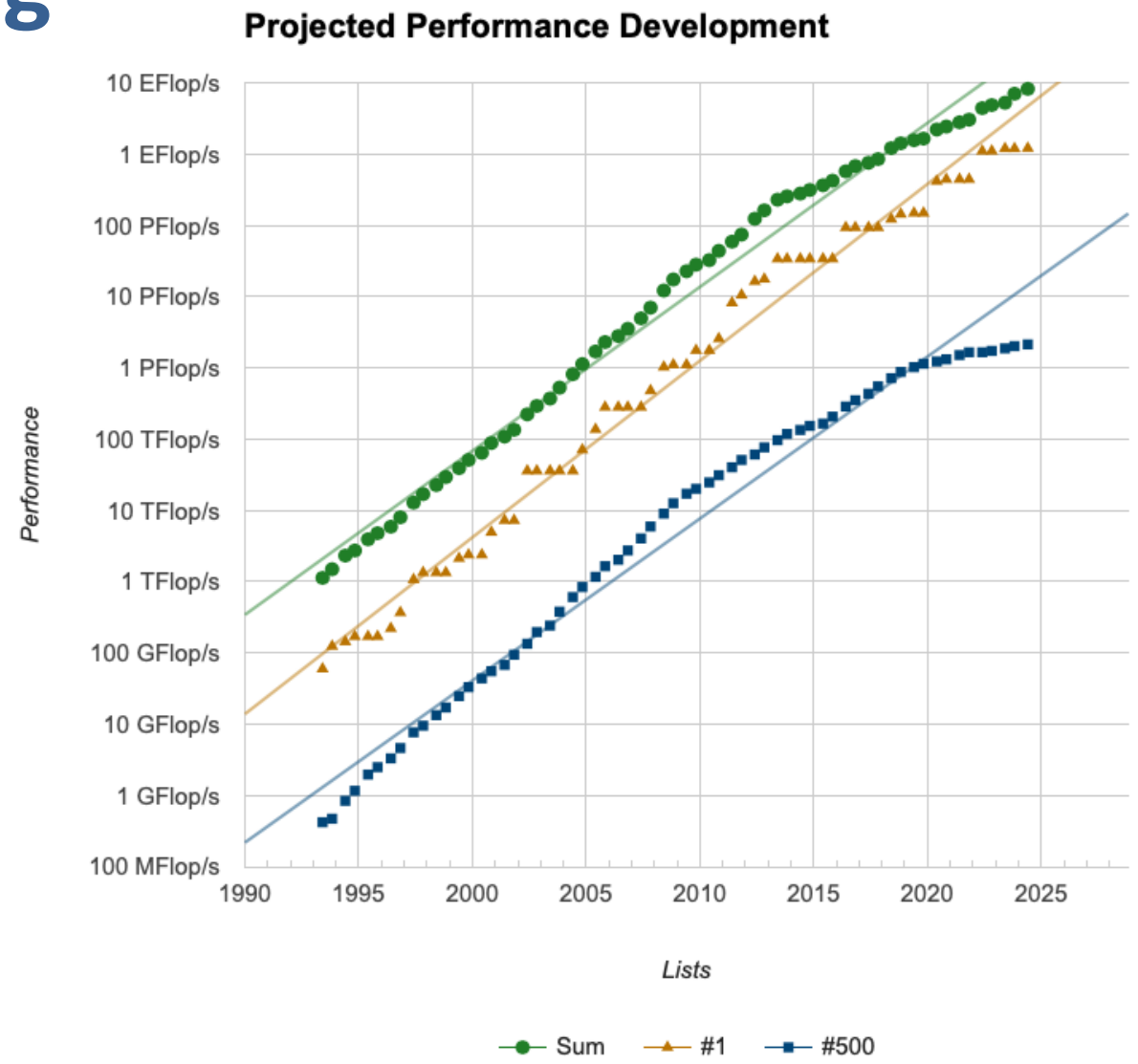
- Procurements
  - *Communicate* datacenter workload to vendors
  - Co-design systems, monitor progress
  - System acceptance (contractual specification)
- Validation of software stack, tools
  - Compilers
  - Debuggers
  - Correctness tools
  - Performance tools
- Research
  - Programming models
  - Computational methods
  - Performance across architectures

Many stakeholders, communication is key for collaboration

Source: https://www.top500.org/statistics/perfdevel/

Benchmarking is challenging

- What are we trying to characterize?
- Are we capturing the best the system can do?
- Is something else impacting performance?
- Did we build and run the code in the *optimal* and *reproducible* way?



HPC benchmarks run on diverse HPC systems

Lawrence Berkeley Nat'l Lab
AMD Zen + **NVIDIA**

F u g a k u
RIKEN **Fujitsu** ARM a64fx

Lawrence Livermore Nat'l Lab
IBM Power9 + **NVIDIA**

Lawrence Livermore Nat'l Lab
AMD Zen + **Radeon**

Oak Ridge National Lab
AMD Zen + **Radeon**

Argonne National Lab
Intel Xeon + Xe

- Benchmark source code
  - Abstraction (OpenMP, RAJA, Kokkos)
  - Hardware-specific (CUDA, ROCm)
- Optimized code for the CPU and GPU
  - Must make effective use of the hardware
  - Can make 10-100x performance difference
- Rely heavily on system packages
  - Need to use optimized communication and MPI libraries that come with machines

Writing benchmark source code is only the beginning

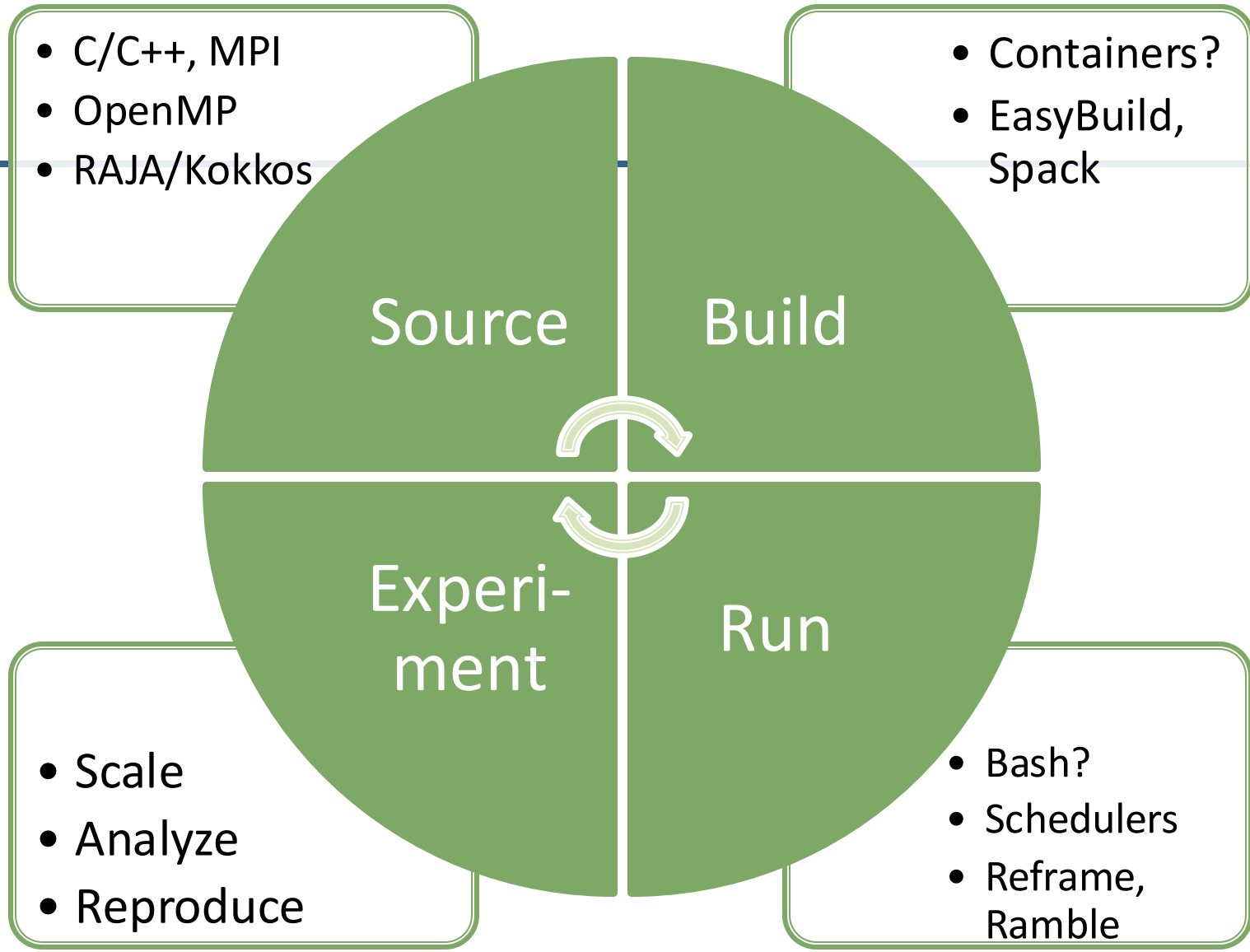
State of the practice:
HPC system benchmarking is manual!

- **Building** on each system is different, porting the builds to new systems is manual
- **Running** on each system is different, porting run scripts to new systems is manual
- Systems keep **changing**, requiring updates to how we build and run benchmarks
- **Triggerinåçg** builds and runs is manual: benchmark results don't stay up to date
- **Performance analysis** of results is manual

Communicate technical specification via code, documentation, white papers

HPC Benchmarks are HPC Software

- Portability
- Maintenance
- Testing/CI
- Verification
- Reproducibility



All components must work *for your system*, focus on explainable *performance*

Benchpark enables complete specification of HPC benchmarks

Infrastructure-as-code benchmark specification codifies:

- Benchmark build and run instructions
- **HPC Systems**
- **HPC Experiments**

github.com/llnl/benchpark

Leverage advances in HPC automation

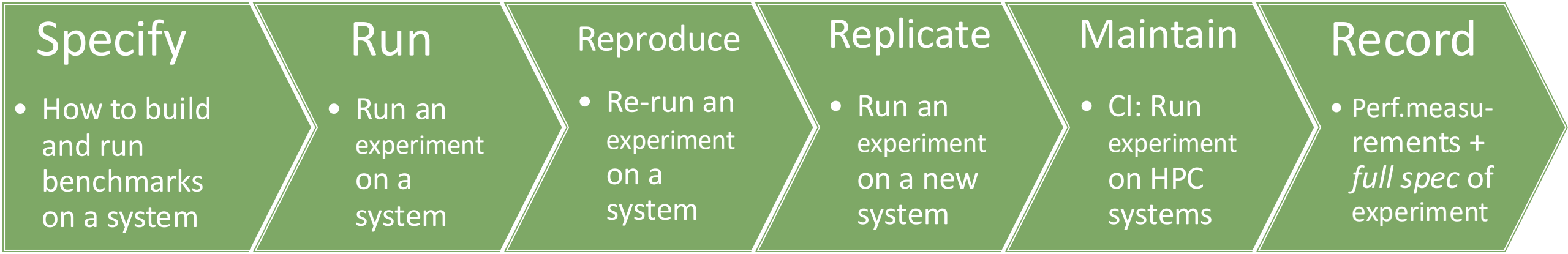
- Source code
- Build specification
- Run specification
- CI

Ramble

Spack

Every part of the specification codified: use to communicate, automate

Benchpark enables *reproducible* specifications of benchmarks



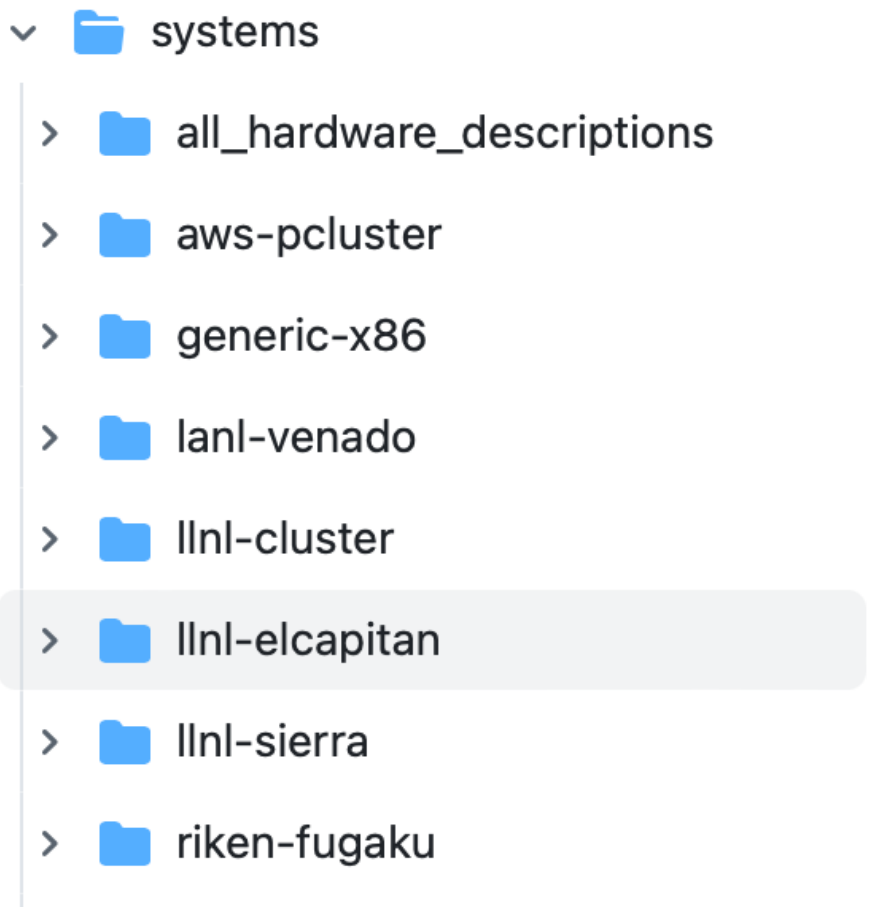
Full specification enables reproducibility, replicability, and automation

HPC System definition for *Performance*

- Resources (CPU cores, GPUs)
- Scheduler (Slurm, flux)
- Process mapping (cores, sockets, GPUs, NICs)
- On-node parallelism (CUDA, ROCM, OpenMP)
- Software stack
  - Compilers (which is best?)
  - MPI implementations (best?)
  - Math libraries

Goal: Use the system correctly

HPC System in Benchpark: *Specify Once*




```
class LlnlElcapitan(System):
    variant(
        "rocm",
        default="5.5.1",
        values=("5.4.3", "5.5.1", ”6.2.4"),
        description="ROCm version",
    )

    variant(
        "compiler",
        default="cce",
        values=("gcc", "cce", "rocmcc"),
        description="Which compiler to use",
    )

    def initialize(self):
        super().initialize()
        self.scheduler = "flux"
        self.sys_cores_per_node = "128"
```

benchpark system init --dest=elcap llnl-elcapitan rocm=6.2.4 compiler=cce

System Specifications

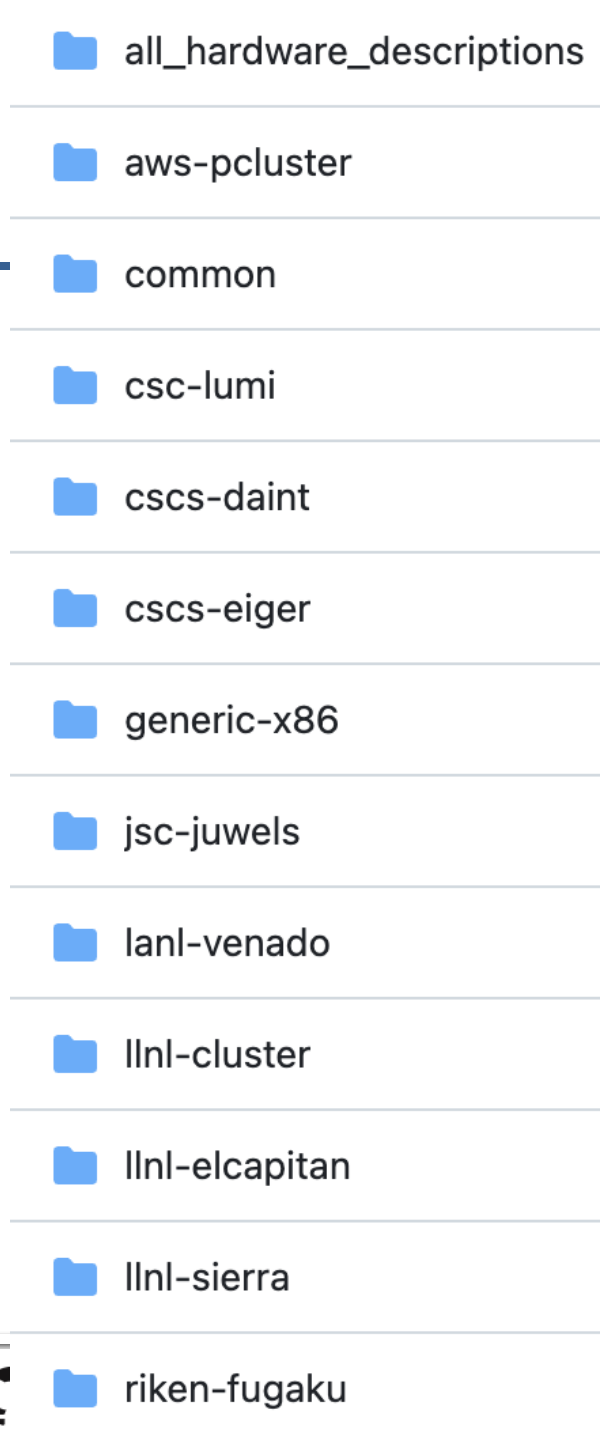
The table below provides a directory of information for systems that have been specified in Benchpark. The column headers in the table below are available for use as the `system` parameter in `benchpark setup`.

Search:

name	integrator.vendor	integrator.name	processor.vendor	processor.name	processor.ISA	processor.u
Atos-zen2-A100-Infiniband	Atos	XH2000	AMD	EPYC-Zen2	x86_64	zen2
AWS_PCluster-zen-EFA	AWS	ParallelCluster	AMD	EPYC-Zen	x86_64	zen
DELL-cascadelake-InfiniBand	DELL		Intel	Xeon6248R	x86_64	cascadelake
DELL-sapphirerapids-OmniPath	DELLEMC	PowerEdge	Intel	XeonPlatinum8480	x86_64	sapphirerapi
Fujitsu-A64FX-TofuD	Fujitsu	FX1000	Fujitsu	A64FX	Armv8.2-A-SVE	aarch64
HPECray-haswell-P100-Infiniband	HPECray		Intel	Xeon-E5-2650v3	x86_64	haswell
HPECray-neoverse-H100-Slingshot	HPECray	EX254n	NVIDIA	Grace	Armv9	neoverse
HPECray-zen2-Slingshot	HPECray		AMD	EPYC-7742	x86_64	zen2
HPECray-zen3-MI250X-Slingshot	HPECray	EX235a	AMD	EPYC-Zen3	x86_64	zen3
HPECray-zen4-MI300A-Slingshot	HPECray	EX255a	AMD	EPYC-Zen4	x86_64	zen4
IBM-power9-V100-Infiniband	IBM	AC922	IBM	POWER9	ppc64le	power9
Penguin-icelake-OmniPath	PenguinComputing	RelionCluster	Intel	XeonPlatinum924248C	x86_64	icelake
Supermicro-icelake-OmniPath	Supermicro		Intel	XeonPlatinum8276L	x86_64	icelake
x86_64					x86_64	

Systems in Benchpark: July 2025

- 4 in Europe
- 6 in US labs
- 1 in Japan
- 1 at a university
- 2 Cloud systems



HPC Experiment definition for *Performance*

- On-node parallelism (CUDA, ROCM, OpenMP)
- Problem sizes
  - Overall problem size, or
  - Per node or per GPU
- Scaling studies
  - How to scale
  - How to decompose
- Resources (cores, GPUs)

Goal: Specify reproducible sets of experiments that map onto specific Systems

HPC Experiment in Benchpark: *Specify Once*

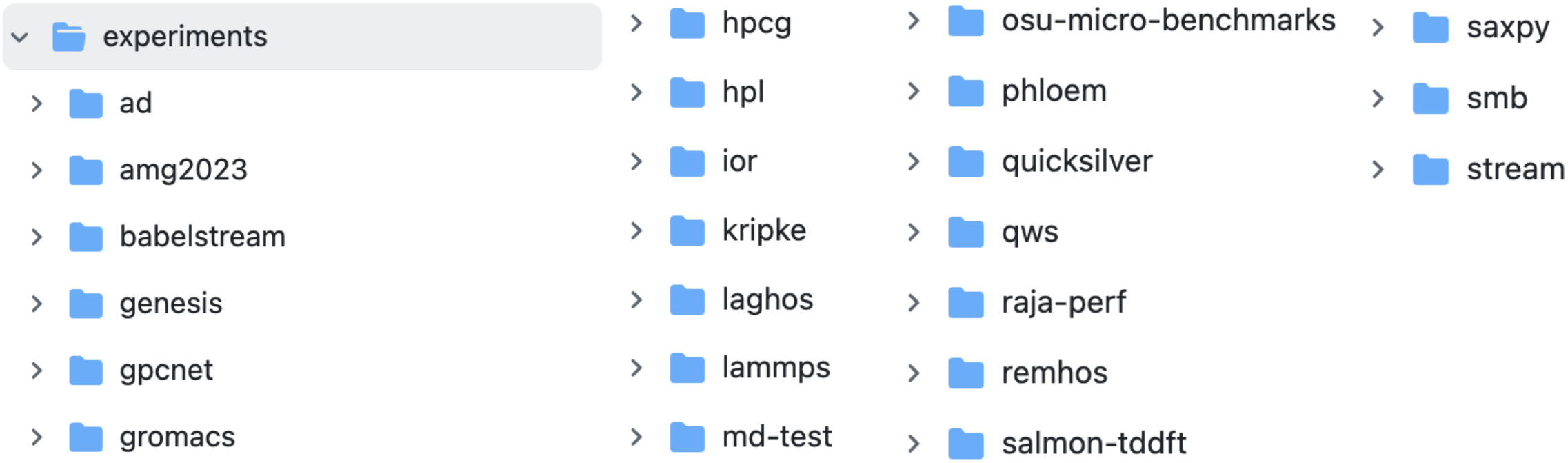
```
class Amg2023(Experiment, OpenMPExperiment, CudaExperiment, ROCmExperiment,
        Scaling(ScalingMode.Strong, ScalingMode.Weak, ScalingMode.Throughput,
                Caliper):
  variant(
    "workload",
    default="problem1",
    values=("problem1", "problem2"),
    description="problem1 or problem2",
  )

  def compute_applications_section(self):
        self.register_scaling_config({
            ScalingMode.Strong: {
                “n_resources”: lambda var, itr, dim, scaling_factor: var.val(dim) * scale,
                “problem_size”: lambda var, itr, dim, scaling_factor: var.val(dim) // scale,
            },
            ScalingMode.Weak: {
                “n_resources”: lambda var, itr, dim, scaling_factor: var.val(dim) * scale,
                “problem_size”: lambda var, itr, dim, scaling_factor: var.val(dim),
            },
            ScalingMode.Throughput: {
                “n_resources”: lambda var, itr, dim, scaling_factor: var.val(dim),
                “problem_size”: lambda var, itr, dim, scaling_factor: var.val(dim) * scale,
```

amg2023 +rocm scaling=strong workload=problem2 caliper=mpi,time

Benchmarks and Experiments

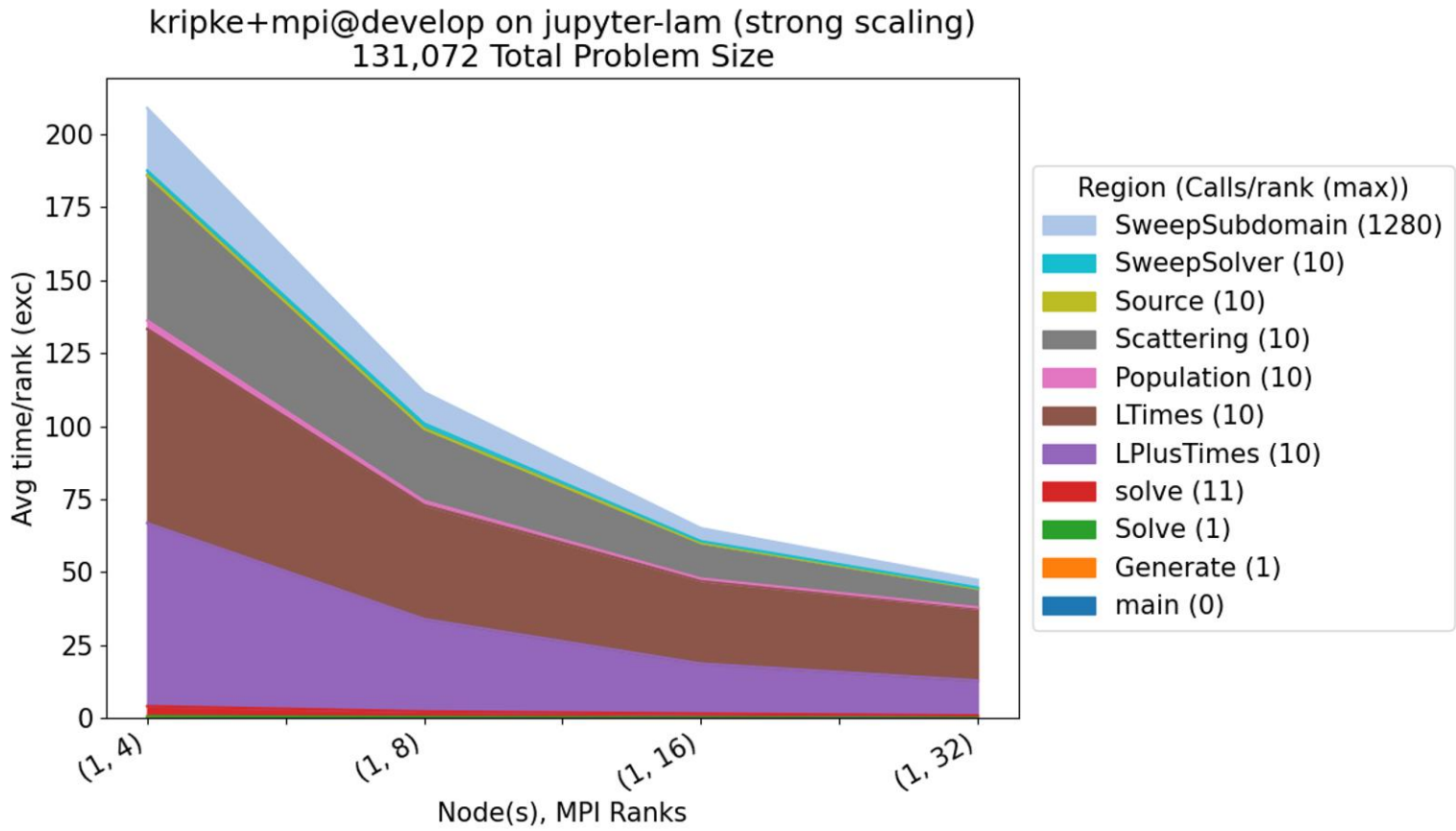
	ad	amg2023
application-domain	[]	['engineering']
benchmark-scale	[]	['large-scale', 'multi-node', 'single-node', 'sub-node']
communication	['mpi']	['mpi']
communication-performance-characteristics	[]	['network-collectives', 'network-latency-bound']
compute-performance-characteristics	[]	['high-branching', 'mixed-precision']
math-libraries	[]	['hypre']
memory-access-characteristics	[]	['high-memory-bandwidth', 'irregular-memory-access', 'large-memory-footprint', 'regular-m
mesh-representation	[]	['block-structured-grid']
method-type	['compiler-transformation']	['solver', 'sparse-linear-algebra']
programming-language	['c', 'c++']	['c']
programming-model	[]	['openmp', 'cuda', 'rocm']
instrumented-caliper	False	True
scaling-experiments	[]	['strong', 'weak']

Experiments in Benchpark: July 2025



- HPL, HPCG
- 4 microbenchmarks
- 3 MPI benchmarks
- 8 US
- 1 Europe
- 2 Japan

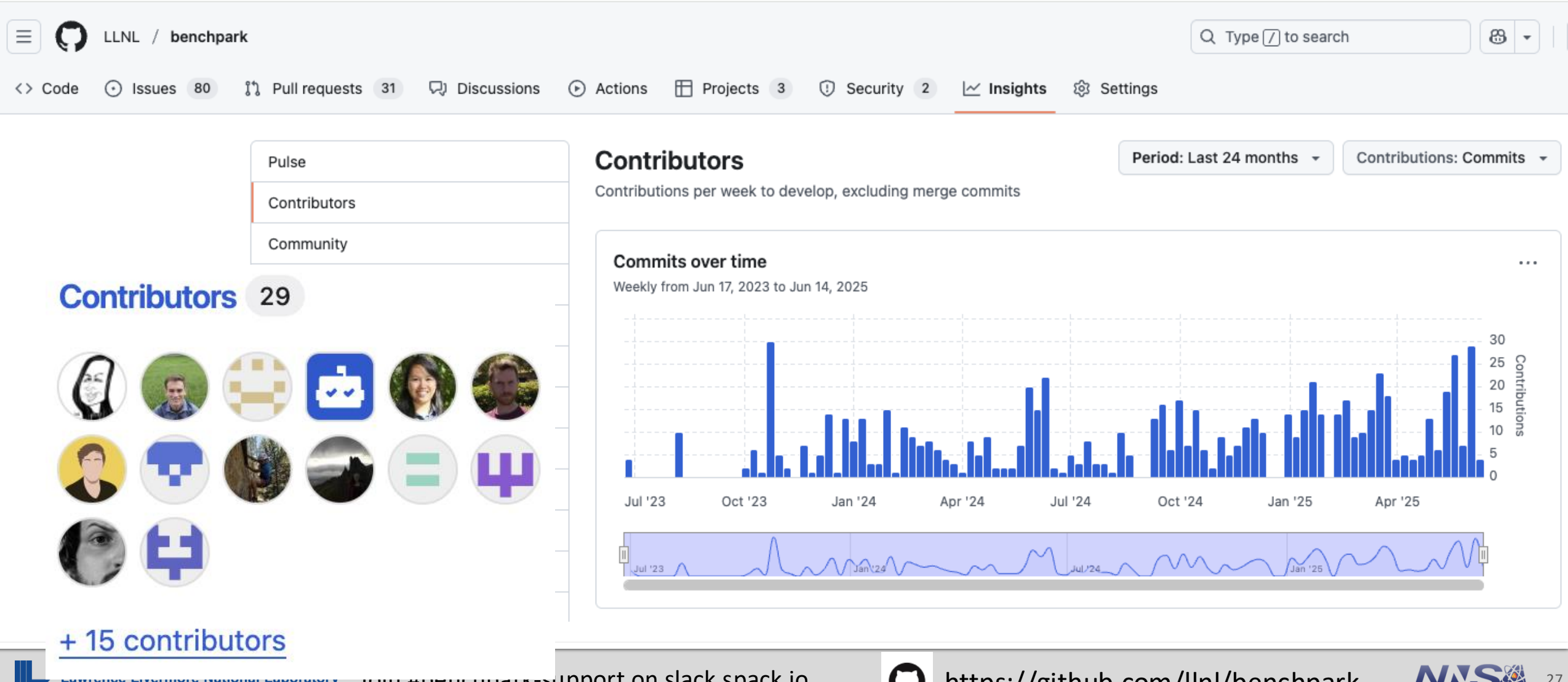
`benchpark analyze` for generating pre-defined analysis charts




```
main
├─ Generate
│  ├─ MPI_Allreduce
│  ├─ MPI_Comm_split
│  └─ MPI_Scan
├─ MPI_Allreduce
├─ MPI_Bcast
├─ MPI_Comm_dup
├─ MPI_Comm_free
├─ MPI_Comm_split
├─ MPI_Finalize
├─ MPI_Finalized
├─ MPI_Gather
├─ MPI_Get_library_version
├─ MPI_Initialized
└─ Solve
   └─ solve
      ├─ LPlusTimes
      ├─ LTimes
      ├─ Population
      │  └─ MPI_Allreduce
      ├─ Scattering
      ├─ Source
      └─ SweepSolver
         ├─ MPI_Irecv
         ├─ MPI_Isend
         ├─ MPI_Testany
         ├─ MPI_Waitall
         └─ SweepSubdomain
```

See https://software.llnl.gov/benchpark/benchpark-analyze.html

Contributions from 11 orgs (60% non-LLNL)



Benchpark codifies benchmarking steps

- Benchpark does **not** replace benchmark source, build system, or Spack package
- Benchpark manages benchmark **experiments** and how they map onto **systems** with specified (or default)
  - Compilers
  - MPI/comm libraries
  - Math libraries
- Start running benchmark experiments on your system with just a few commands

```
git clone git@github.com:LLNL/benchpark.git
```

```
benchpark list systems
benchpark system init --dest=elcap llnl-elcapitan
```

```
benchpark list experiments
benchpark experiment init dest=amg
                          amg2023+rocm+strong
```

```
benchpark setup amg elcap workspace
ramble workspace setup
```

```
ramble on
```

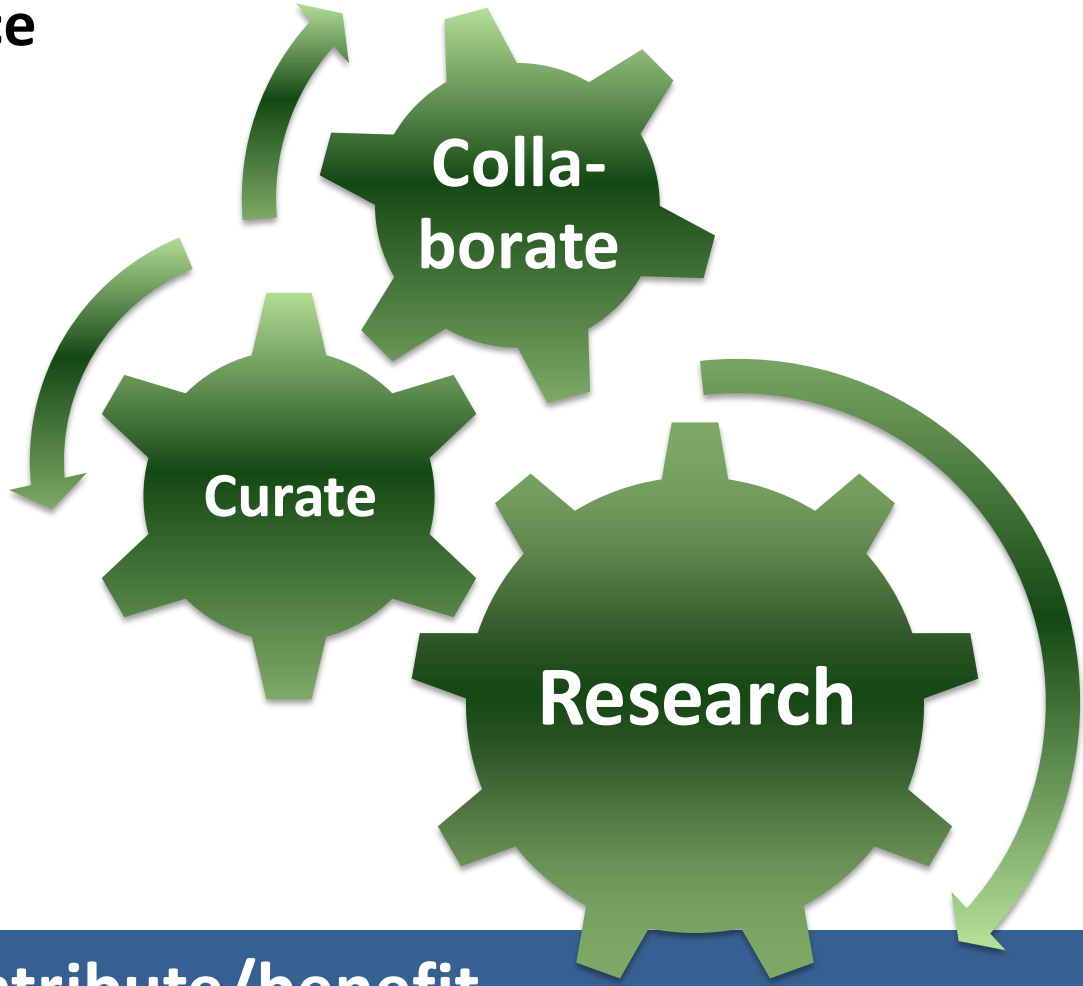
Who can use Benchpark

People who want to use or distribute benchmarks for HPC!

1. **End Users of HPC Benchmarks**
   - Install, run, analyze performance of HPC benchmarks
2. **Benchmark Developers**
   - People who want to share their benchmarks
3. **Procurement teams at HPC Centers**
   - Curate workload representation, evaluate and monitor system progress
4. **HPC Vendors**
   - Understand the curated workload of HPC centers, propose systems

Catalogued library of *working* benchmarks

- **Enables exploration of large configuration space**
  - —Architectural
  - —Software stack
  - —Temporal
- What architecture and system configuration is best for *my benchmark*?
- On *my system*, is OpenMPI good enough?
- What purpose will Benchpark help you address?



Building a community to contribute/benefit

Benchpark: Open collaborative repository for *reproducible* specifications of HPC benchmarks

Infrastructure-as-code benchmark specification enables **reproducibility, replicability, and automation**

- HPC Systems
- HPC Experiments

Specify	Run	Reproduce	Replicate	Maintain	Record
• How to build and run benchmarks on a system	• Run an experiment on a system	• Re-run an experiment on a system	• Run an experiment on a new system	• CI: Run experiment on HPC systems	• Perf.measu-rements + *full spec* of experiment

- Tagging, keywords for publications
- Performance metrics, metrics of usefulness
- Dashboards: Archive specs+results

- CI pipelines on PRs from GitHub at data centers across the world and in the cloud

*Olga Pearce et al, Continuous Benchmarking, HPCTests SC|23

Full specification enables reproducibility, replicability, and automation

Benchpark roadmap and community engagement

Future directions:

- Suite curation: Reproducible specification of an entire suite
- Tagging: Keywords for publications, finding benchmarks
- Metrics: Performance, usefulness
- Dashboards: Archive+share *specs*+results, Slices in configuration space
- CI pipelines at data centers across the world and in the cloud

Community Engagement:

- Co-design / vendors on board, but incentive for app teams? (carrot or stick?)
- Who owns which parts of the specification and approves changes?
- Who finances R&D and maintenance?
- ROI for the people working on it? → think about post docs, researchers, etc.

Collaboration, reproducibility, fully specified public results

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- All tutorial materials
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  - software.llnl.gov/benchpark/example-workflow.html
- Connect with us on Spack slack

  #benchpark-support

- We want your feedback!

Thank you!

slack.spack.io
#benchpark-support

★ Star us on GitHub!

github.com/llnl/caliper

github.com/llnl/thicket

github.com/llnl/benchpark

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